



C/N_0 estimation based on acquisition correlation ratio for short GNSS data

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Abstract

Carrier-to-noise ratio (C/N_0) represents the ratio of signal power and noise power density, and it is of great significance in many applications of Global Navigation Satellite System (GNSS), such as satellite signal quality monitoring, spoofing detection, position accuracy estimation, etc. Accurate C/N_0 estimation results will benefit a lot for short data situations like snapshot receivers, fast spoofing detection, adjusting tracking loop parameters based on fast C/N_0 estimation and so on. However, most classical C/N_0 estimation methods rely on the tracking stage of a GNSS receiver, which requires at least seconds of data to get an accurate estimation and further limits the application of C/N_0 for such situations. To address the limitations of C/N_0 estimation for short GNSS data, we introduce a C/N_0 estimation method based on the Acquisition Correlation Ratio (ACR) that estimates C/N_0 within only a few milliseconds signals. We derive the theoretical performance of ACR and three classical C/N_0 estimation methods. It can be proved that the proposed ACR method has lower theoretical RMSE than the other three methods. Monte Carlo simulations are consistent with the theoretical analysis, which indicates that the accuracy of ACR method can approach the theoretical bound. Experiment results of signal simulator test and real-world test both indicate that the proposed method can enhance the estimation accuracy of C/N_0 under short data conditions, achieving an estimation RMSE improvement of 0.8 dB-Hz on average with 20 ms data.

Keywords GNSS · C/N_0 estimation · Acquisition correlation value · Short data

Introduction

Carrier-to-noise ratio (C/N_0), which represents the ratio of signal power to noise power within a unit bandwidth, is widely used in the field of GNSS. Measurements of C/N_0 are not only the extra information of receiver, but also the important statistics which are used in many applications for receiver tracking loop performance prediction (Issa et al. 2021), weighted least squares positioning (Kaplan and Hegarty 2017), mitigation of carrier phase multiple errors (Rost and Wanninger 2009), spoofing detection based on the temporal and spatial feature of C/N_0 (Dehghanian et al. 2012; Kraemer et al. 2012), etc.

Many C/N_0 estimation methods have been proposed so far. The Narrowband-Wideband Power Ratio (NWPR) method is one of the most widely used method which usually

serves as the benchmark due to its high estimation accuracy (Sayre 2003). The Correlator Comparison Method (CCM) has low complexity but require additional correlators (Groves 2005). The Moment Method (MM) is easy to compute but have unsatisfactory accuracy for low C/N_0 (Sharawi et al. 2007). Real Signal Complex Noise (RSCN) method and Squared Signal-to-Noise Variance (SSNV) method are straightforward but inaccurate (Falletti et al. 2011). Additionally, the Extended Kalman Filter (EKF) has also been applied to improve estimation rate for one-bit quantization receivers (Issa et al. 2021).

All the starting points of aforementioned C/N_0 estimation methods are In-phase (I) and Quadrature (Q) integration values of GNSS receiver tracking loops. But the tracking loops require continuous adjustment to converge to stable states with high accuracy due to the bias of the initialization parameters. Considering the reliance on stable tracking and the positive relationship between estimation accuracy and data length, these methods require at least seconds of data to obtain accurate estimation results (Sharawi et al. 2007), which makes these methods not applicable for short GNSS data.

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The C/N_0 estimation of snapshot GNSS receivers is one of the most typical limited cases. Snapshot receivers are small, light and low-power GNSS tracking devices (Luo et al. 2023), which usually capture only 10 ms to 100 ms signals for each position computation with the help of ephemerides and a rough knowledge of the position (Gill and Akos 2023; Wang et al. 2019). C/N_0 can be combined with the position algorithm as the prior information to improve the accuracy (Fernández Hernández and Borre 2016). In addition, C/N_0 estimation with short GNSS data is of great significance for compressing the data transmission volume of GNSS signal monitoring (Spacek and Kovar 2007). Besides, the situations where C/N_0 estimation under short data conditions matters include fast spoofing detection utilizing C/N_0 information during the acquisition stage (Zhang et al. 2020; Jahromi et al. 2012), adaptively adjusting the tracking loop parameters based on fast C/N_0 estimation during the initialization stage to accelerate convergence (Cheng and Chang 2020), detection and mitigating the impact of multipath error by comparing the C/N_0 estimation results of different stages (Kubo et al. 2020) and so on. In such situations, receivers either do not need the tracking loop or cannot output accurate integration values for C/N_0 estimation due to the short length of data. So far, almost no scholars have proposed effective methods to solve the C/N_0 estimation problem under short data condition.

In this paper, we first evaluate three classical C/N_0 estimation methods, namely NWPR, CCM and MM under short data conditions. Theoretical and simulation analysis shows that the performance of the three classical methods degrades and cannot satisfy the accuracy requirements of C/N_0 in short data situations. To address the limitation of classical methods under such conditions, we propose a C/N_0 estimation method based on the ratio of acquisition correlation values, which is called the Acquisition Correlation Ratio (ACR) method. The basic ACR method and confirmatory experiment results were first introduced in our recent conference paper (Ma et al. 2023). In this paper, we establish more comprehensive hypothesis models and analyze the theoretical performance of ACR method and three classical methods, together with detailed simulation and experimental results of signal simulator and real-world tests, which are especially designed for this paper.

Based on the fact that the signal and noise power have different effects on different acquisition correlation values, the ACR method constructs the ratio of acquisition correlation values corresponding to the same frequency but different code phases and further establishes its relationship with C/N_0 . In addition, the ACR method only requires millisecond-level data to estimate C/N_0 . Both theoretical analysis and simulation results show that the ACR method can provide a more accurate C/N_0 estimation than other methods

under short data conditions. Experimental results of signal simulator and real-world tests also validate the improvement of ACR. Compared with classical methods, the ACR method fully utilizes the auto-correlation function features of Pseudo Random Noise (PRN) codes rather than only focusing on the prompt I and Q correlation values in the tracking loop, which leads to better performance.

The subsequent sections are organized as follows. The GNSS signal processing and C/N_0 calculation models are presented in the second section. The third section presents the performance analysis of three classical methods, and the fourth section elaborates the proposed ACR method. The fifth section derives the theoretical performance and conducts simulations. The six section presents the experiment results, followed by the conclusions in the last section.

Signal processing and C/N_0 calculation model

The section briefly reviews the models for correlation values and C/N_0 calculation (Ma et al. 2023). Taking the traditional GNSS signal as an example, the intermediate frequency signal of m th satellite processed by a GNSS receiver can be expressed as (Sharawi et al. 2007)

$$s^{(m)}(t_n) = A^{(m)}D^{(m)}(t_n - \tau_a^{(m)})P^{(m)}(t_n - \tau_a^{(m)}) \cos \left(2\pi \left(f_i + f_d^{(m)} \right) t_n + \theta^{(m)} \right) + w^{(m)}(t_n) \quad (1)$$

where t_n denotes the sampling time and f_s denotes the sampling rate. A , $D(\cdot)$ and $P(\cdot)$ denote the signal amplitude, navigation data and PRN code respectively. τ_a denotes the code phase, f_i denotes the intermediate frequency of the signal, f_d denotes the Doppler frequency, θ denotes the initial carrier phase, and $w(\cdot)$ denotes the noise. Multiplied by local quadrature carriers, the signal can be transformed to I and Q signals as expressed below, where the satellite number m is omitted (Ma et al. 2023)

$$s_I(t_n; f) = AD(t_n - \tau_a)P(t_n - \tau_a) \cos(2\pi f_e + \theta_e) + w_I(t_n) \quad (2)$$

$$s_Q(t_n; f) = AD(t_n - \tau_a)P(t_n - \tau_a) \sin(2\pi f_e + \theta_e) + w_Q(t_n) \quad (3)$$

where f denotes the local carrier frequency, $f_e = f - (f_i + f_d)$ and θ_e denote the differences of carrier frequency and initial carrier phase. Respectively, $w_I(t_n)$ and $w_Q(t_n)$ denote the I and Q noise. Generally, I and Q noise are assumed to be independent and obey the zero-mean Gaussian distribution $\mathcal{N}(0, \sigma_0^2)$ where σ_0^2 is the variance.

Representing the normalized PRN code auto-correlation function with $R(\cdot)$ and assuming $D(\cdot) = 1$, the I and Q

integration values can be obtained by correlating $s_I(t_n; f)$ and $s_Q(t_n; f)$ with the local PRN code (Ma et al. 2023)

$$I_i(f, \tau) = AR(\tau_e) \sin c(f_e T_{\text{coh}}) \cos(\phi_i) + w_{I,i} \quad (4)$$

$$Q_i(f, \tau) = AR(\tau_e) \text{sinc}(f_e T_{\text{coh}}) \sin(\phi_i) + w_{Q,i} \quad (5)$$

where i denotes the index of integration values, τ denotes the local code phase, $\tau_e = \tau_a - \tau$ and $\phi_i = \pi f_e (\hat{t}_i + T_{\text{coh}}) + \theta_e$. $\hat{t}_i$ denotes the initial time of integration. $w_{I,i}$ and $w_{Q,i}$ obey the Gaussian distribution $\mathcal{N}(0, \sigma^2)$, in which $\sigma_0^2 = N\sigma^2$. $N = f_s T_{\text{coh}}$ denotes the number of sampling points in the coherent integration time T_{coh} . Local PRN code and carrier will be aligned with the received signal when the tracking loops converge ideally, which means $\tau_e = 0$, $f_e = 0$ and $\theta_e = 0$. Then, the above equations can be further simplified as

$$I_i(f, \tau) = A + w_{I,i} \quad Q_i(f, \tau) = w_{Q,i} \quad (6)$$

According to the definition of C/N_0 (Kaplan and Hegarty 2017), the logarithmic form of C/N_0 in (2) and (3) without considering implementation loss and receiver hardware limitations can be expressed as follows

$$\begin{aligned} C/N_0(\text{dB} - \text{Hz}) &= 10 \lg \left(\frac{A^2}{2\sigma_0^2} f_s \right) = 10 \lg \left(\frac{A^2}{2N\sigma^2} f_s \right) \\ &= 10 \lg \left(\frac{A^2}{2\sigma^2 T_{\text{coh}}} \right) \end{aligned} \quad (7)$$

Equations (4) to (7) matter a lot as the basis of C/N_0 estimation.

Review of classical C/N_0 estimation methods and performance analysis

This section elaborates three classical C/N_0 estimation methods and derives theoretical performance. In addition, we conduct simulations under different data length conditions. The results show the performance degeneration of classical methods under short data conditions and the necessity to come up with a new method to accurately estimate C/N_0 for short GNSS data.

Narrowband-wideband power ratio method

The NWPR method constructs an estimator of C/N_0 by comparing the narrowband power (NBP) and the wideband power (WBP). The k th estimation of WBP and NBP can be expressed as follows (Muthuraman and Borio 2010)

$$\text{NBP}_k = \left(\sum_{i=1}^M I_i \right)^2 + \left(\sum_{i=1}^M Q_i \right)^2 \quad \text{WBP}_k = \sum_{i=1}^M (I_i^2 + Q_i^2) \quad (8)$$

where I_i and Q_i represent the prompt correlator values, and M denotes the number of coherent integration values. The bandwidth of NBP is $1/MT_{\text{coh}}$, while the bandwidth of WBP is $1/T_{\text{coh}}$. The power ratio and its average of K measurements are defined as follows

$$\text{NP}_k = \frac{\text{NBP}_k}{\text{WBP}_k} \quad \overline{\text{NP}} = \frac{1}{K} \sum_{k=1}^K \text{NP}_k \quad (9)$$

Further the C/N_0 estimator can be constructed

$$(C/N_0)_N = \frac{1}{T_{\text{coh}}} \frac{\overline{\text{NP}} - 1}{M - \overline{\text{NP}}} \quad (10)$$

where the subscript N denotes NWPR. Its theoretical performance can be derived as (Groves 2005)

$$E[(C/N_0)_N] \approx C/N_0 + \left(C/N_0 + \frac{\left(\frac{A^2}{2\sigma^2} + 1 \right)^2}{(M^2 - 1)T_{\text{coh}}} \right) \cdot \frac{(M^2 - M) \left[\left(\frac{A^2}{2\sigma^2} \right)^2 + \frac{A^2}{\sigma^2} \right] + M - 1}{KM \left(\frac{A^2}{2\sigma^2} + 1 \right)^4} \quad (11)$$

$$\text{Var}[(C/N_0)_N] \approx \frac{1}{T_{\text{coh}}^2} \frac{(M-1)^2}{\left[M - \frac{\frac{A^2}{2\sigma^2} M + 1}{\frac{A^2}{2\sigma^2} + 1} + \frac{\frac{A^2}{2\sigma^2} (1-M)}{M \left(\frac{A^2}{2\sigma^2} + 1 \right)^2} \right]^4} \cdot \frac{(M^2 - M) \left[\left(\frac{A^2}{2\sigma^2} \right)^2 + \frac{A^2}{\sigma^2} \right] + M - 1}{KM \left(\frac{A^2}{2\sigma^2} + 1 \right)^4} \quad (12)$$

which show that the C/N_0 estimator of NWPR is biased.

Correlator comparison method

The CCM estimates C/N_0 by comparing the prompt correlator outputs I_p and Q_p with the asynchronous correlator output I_A which represents the power of the noise channel. The variable Z is defined as follows (Xie et al. 2009)

$$Z = \frac{\sum_{i=1}^N (I_{Pi}^2 + Q_{Pi}^2)}{2 \sum_{i=1}^N I_{Ai}^2} \quad (13)$$

Then the estimator can be constructed as follows

$$(C/N_0)_C = \frac{1}{T_{\text{coh}}} \left(\frac{N}{N+2} Z - 1 \right) \quad (14)$$

where the subscript C denotes CCM. Its theoretical performance can be derived as (Groves 2005)

$$E[(C/N_0)_C] \approx C/N_0 \quad \text{Var}[(C/N_0)_C] \approx \frac{N \left[2 \left(\frac{A^2}{2\sigma^2} \right)^2 + 6 \frac{A^2}{2\sigma^2} + 3 \right]}{(N+2)^2 T_{\text{coh}}^2} \quad (15)$$

Equation (15) represents the unbiased properties of CCM estimator.

Moment method

The Moment Method (MM) utilizes the second-order moment M_2 and fourth-order moment M_4 of the prompt I and Q integration values to estimate C/N_0 , which can be expressed as follows

$$M_2 = \frac{1}{M} \sum_{i=1}^M (I_i^2 + Q_i^2) \quad M_4 = \frac{1}{M} \sum_{i=1}^M (I_i^2 + Q_i^2)^2 \quad (16)$$

Then, the estimator can be constructed as follows (Falletti et al. 2011)

$$\text{Var}[(C/N_0)_M] \approx \frac{1}{MT_{\text{coh}}^2} \left[\left(\frac{A^2}{\sigma^2} \right)^3 + \frac{33}{2} \left(\frac{A^2}{\sigma^2} \right)^2 + 92 \frac{A^2}{\sigma^2} + 210 + 204 \left(\frac{A^2}{\sigma^2} \right)^{-1} + 68 \left(\frac{A^2}{\sigma^2} \right)^{-2} \right] \quad (21)$$

$$(C/N_0)_M = \frac{1}{T_{\text{coh}}} \frac{\sqrt{2M_2^2 - M_4}}{M_2 - \sqrt{2M_2^2 - M_4}} \quad (17)$$

where the subscript M denotes MM. Owing to its complex expression, it is difficult to determine the performance evaluation using the standard formulas for random

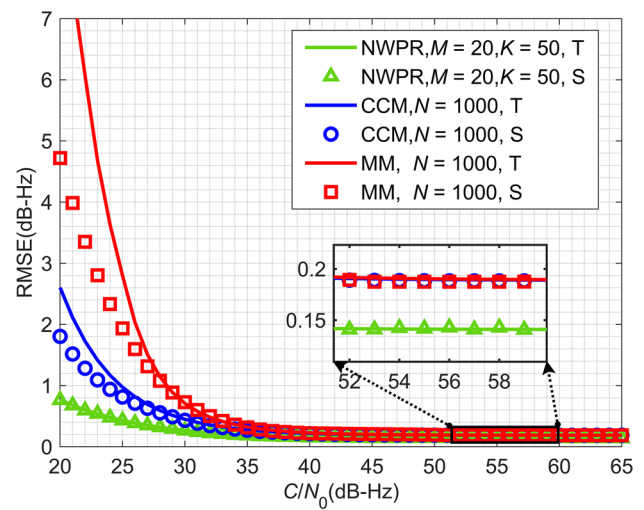


Fig. 1 Comparison of C/N_0 estimation RMSE for three classical methods with 1000 ms data and $T_{\text{coh}} = 1$ ms. The symbol “T” in the legend denotes “Theory” and “S” denotes “Simulation” which are the same for other figures in the paper

variable transforms. One method is to approximate through Taylor expansion. For the estimator $\theta = g(\mathbf{T})$, where $\mathbf{T} = [T_1, T_2, \dots, T_r]$, the Taylor expansion is performed near $\mathbf{T} = E(\mathbf{T}) = \boldsymbol{\mu}$ (Kay 1993)

$$\theta = g(\mathbf{T}) \approx g(\boldsymbol{\mu}) + \sum_{k=1}^r \left. \frac{\partial g}{\partial T_k} \right|_{\mathbf{T}=\boldsymbol{\mu}} (T_k - \mu_k) \quad (18)$$

Then, its expectation and variance can be expressed

$$E(\theta) \approx g(\boldsymbol{\mu}) \quad \text{Var}(\theta) \approx \left. \frac{\partial g}{\partial \mathbf{T}} \right|_{\mathbf{T}=\boldsymbol{\mu}}^T \mathbf{C}_T \left. \frac{\partial g}{\partial \mathbf{T}} \right|_{\mathbf{T}=\boldsymbol{\mu}} \quad (19)$$

where $\mathbf{C}_T$ denotes the covariance matrix of $\mathbf{T}$. By applying the formula, the theoretical performance of MM can be obtained that

$$E[(C/N_0)_M] \approx C/N_0 \quad (20)$$

where the detailed derivation is shown in appendix. The equations above mean $(C/N_0)_M$ is approximately unbiased.

Summary

To comprehensively measure the bias and variance of estimated results, Root Mean Squared Error (RMSE) is chosen

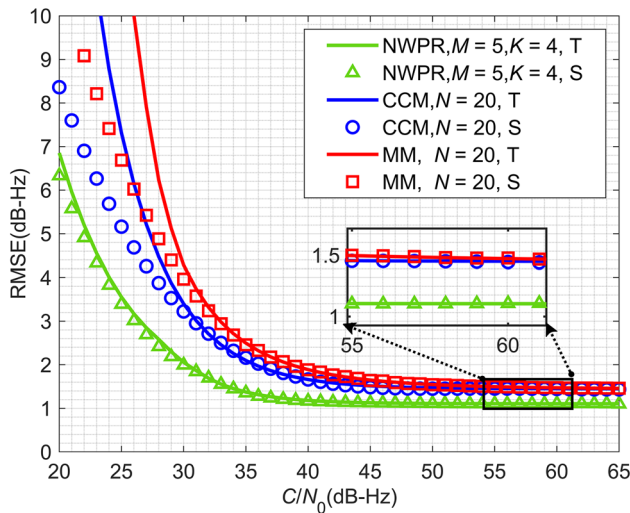


Fig. 2 Comparison of C/N_0 estimation RMSE for three classical methods with 20 ms data and $T_{\text{coh}} = 1$ ms

as the evaluation indicator for comparison. Its definition is as follows (Kay 1993)

$$\text{RMSE} = \sqrt{\frac{1}{K} \sum_{k=1}^K \left[(C/N_0)_{\text{est},k} - (C/N_0)_{\text{true}} \right]^2} \quad (22)$$

where $(C/N_0)_{\text{est}}$ is the estimation, $(C/N_0)_{\text{true}}$ is the real value.

A Monte Carlo simulation with 10,000 runs is conducted on a MATLAB platform with the immediate frequency $f_i = 4.13$ MHz and sampling rate $f_s = 16.367$ MHz. Figure 1 illustrates the theory and simulation RMSE results of different methods under 1,000 ms data when local carrier and code replicas are perfectly synchronized with the received signal. It can be seen that the simulation results fit the theory deduction well and the NWPR method exhibits the best performance.

However, all of these methods result in high RMSE larger than 1 dB-Hz when dealing with short data, as shown in Fig. 2. It can be seen that the simulation results deviate from the theory because the deduction of theoretical performance is approximate in many equations and the approximation accuracy decreases with shorter data or lower C/N_0 (Kay 1993). The difference between theory and simulation results of MM is more obvious owing to the first-order approximation in (18) while the third-order approximation in NWPR and CCM (Shao 2003).

Additionally, in cases where signal is short, tracking loops of a GNSS receiver cannot converge immediately, leading to unstable I and Q integration values which will further deteriorate the RMSE of the estimation results. Therefore, the classical methods fail to meet the requirement of accurate

C/N_0 estimation for short data. The next section will introduce the proposed ACR method, a newly proposed C/N_0 estimation method for short data situations.

Proposed acquisition correlation ratio (ACR) method

Signal acquisition can estimate the carrier frequency and code phase with only a few milliseconds of data. Acquisition correlation values imply the relationship between the ratio of the signal power and noise power. Considering different effects of signal and noise power on different correlation values, the section introduces a C/N_0 estimation method based on the ratio of Acquisition Correlation Ratio, i.e., the ACR method. The core of ACR is to utilize the correlation values corresponding to the same carrier frequency but different code phases to reflect the impact of C/N_0 .

Algorithm description

The subsection describes the process of the ACR method and makes appropriate modifications based on the previous work to make the algorithm more concise (Ma et al. 2023). To ensure that the acquisition result is not affected by the navigation data and initial carrier phase, the acquisition correlation value is constructed as follows (Kaplan and Hegarty 2017)

$$V_k(f, \tau) = \sum_{i=k}^{k+N_c-1} [I_i(f, \tau)^2 + Q_i(f, \tau)^2] \quad (23)$$

where k denotes the index of acquisition correlation values and $N_c = 1, 2, \dots$ denotes the number of noncoherent integration. The global maximum value V_k^m and the corresponding carrier frequency and code phase $(f_{d,k}, \tau_k)$ are defined as

$$V_k^m = \max_{f, \tau} [V_k(f, \tau)] \quad (f_{d,k}, \tau_k) = \arg \max_{f, \tau} [V_k(f, \tau)] \quad (24)$$

V_k^s is defined as the largest correlation value among those values sharing the same carrier frequency as the global maximum value V_k^m but differing by more than 1 chip in code phase from it

$$V_k^s = \max_{|\tau - \tau_k| > 1} [V_k(f_{d,k}, \tau)] \quad \hat{\tau}_k = \arg \max_{\tau} [V_k(f_{d,k}, \tau)] \quad (25)$$

where $\hat{\tau}_k$ denotes the corresponding code phase. Selecting V_k^s with the same frequency as V_k^m simplifies the analysis complexity influenced by Doppler deviations and helps to reduce the impact of satellite multi-access interference by taking advantage of the characteristics of the sinc($\cdot$) function in (4) and (5).

For the global maximum value $V_k^m, (f_{d,k}, \tau_k)$ is close to the actual frequency and code phase (f_a, τ_a) , so it can be approximated that $\text{sinc}(f_e T_{\text{coh}}) \approx 1$, $R(\tau_e) \approx 1$ where $f_e = f_d - f_{d,k}$ and $\tau_e = \tau_a - \tau_k$. The influence of f_e and τ_e will be discussed in next section. Assuming that $R(\tau_a - \hat{\tau}_k) = p_k$ for V_k^s , it can be obtained that

$$\begin{aligned} I_i(f_{d,k}, \tau_k) &= A \cos(\phi_i) + w_{1,i} \\ I_i(f_{d,k}, \hat{\tau}_k) &= p_k A \cos(\phi_i) + w_{1,i} \end{aligned} \quad (26)$$

$$\begin{aligned} Q_i(f_{d,k}, \tau_k) &= A \sin(\phi_i) + w_{Q,i} \\ Q_i(f_{d,k}, \hat{\tau}_k) &= p_k A \sin(\phi_i) + w_{1,i} \end{aligned} \quad (27)$$

For convenience of analysis, V_k^m and V_k^s are normalized by the variance of noise

$$\tilde{V}_k^m = \frac{V_k^m}{\sigma^2} \quad \tilde{V}_k^s = \frac{V_k^s}{\sigma^2} \quad (28)$$

Owing to the mutual independence of I_i and Q_i , $\tilde{V}_k^m = V_k(f_{d,k}, \tau_k)$ obeys the non-central chi-square distribution χ_{2N_c, λ_0}^2 with $2N_c$ degrees of freedom and non-central parameter λ_0 which is shown as follows (Kennedy and Gentile 2021)

$$\lambda_0 = \sum_{i=k}^{k+N_c-1} \left[\frac{A^2}{\sigma^2} \cos^2(\phi_i) + \frac{A^2}{\sigma^2} \sin^2(\phi_i) \right] = 2N_c T_{\text{coh}} \cdot (C/N_0) \quad (29)$$

$\tilde{V}_k^s$ is equivalent to the maximum value of different independent and non-identically distributed chi-square distribution random variables in (25). Assuming the Probability Density Function (PDF) is $f_{\tilde{V}_k^s}(x)$, then it can be obtained that

$$E(\tilde{V}_k^m) = 2N_c + \lambda_0 \quad E(\tilde{V}_k^s) = \int_0^{+\infty} x f_{\tilde{V}_k^s}(x) dx \quad (30)$$

The deduction of $f_{\tilde{V}_k^s}(x)$ will be discussed in detail in next subsection. Taking N measurements of V_k^m and V_k^s to reduce the randomness, the acquisition correlation ratio r_A is defined as

$$r_A = \frac{\sum_{k=1}^N V_k^m / N}{\sum_{k=1}^N V_k^s / N} = \frac{\sum_{k=1}^N \tilde{V}_k^m / N}{\sum_{k=1}^N \tilde{V}_k^s / N} \approx \frac{E(\tilde{V}_k^m)}{E(\tilde{V}_k^s)} \quad (31)$$

Substituting (30) into (31), the relationship between r_A and C/N_0 can be established

$$r_A = g(C/N_0) = \frac{2N_c + \lambda_0}{\int_0^{+\infty} x f_{\tilde{V}_k^s}(x) dx} \quad (32)$$

Therefore, C/N_0 can be estimated because r_A which can be obtained by observation (Ma et al. 2023)

$$(C/N_0)_A = g^{-1}(r_A) \quad (33)$$

where the subscript A denotes ACR. Since $g^{-1}(r_A)$ is difficult to be expressed explicitly, we can estimate C/N_0 with numerical methods in practical applications.

Hypothesis models

In this section, different hypothesis models for $\tilde{V}_k^s$ with different computational complexity and estimation accuracy are proposed. The third-order model considers full features of the pseudo-code auto-correlation function while the second-order and first-order models make some simplified assumptions at the cost of precision. For the concrete analysis, the subsection takes the GPS L1 C/A code signal as an example, which is easy to expand to other satellite navigation signals.

Third-order model

There are three possible values apart from 1 for the normalized auto-correlation function of GPS L1 C/A code: $|p_1| = 65/1023$, $|p_2| = 63/1023$ and $|p_3| \approx 0$, which are non-ideal correlation values due to sequence length limitations (Kaplan and Hegarty 2017). Assume that the number of peaks corresponding to p_n is N_n in auto-correlation function. $\tilde{V}_{k,n}$ belongs to a subset of $\tilde{V}_k = V_k/\sigma^2$ and it represents the auto-correlation value with noise corresponding to p_n where $n = 1, 2, 3$. $\mathcal{K}_n$ denotes the set of possible k corresponding to p_n and the maximum value $\tilde{V}_{k,n}^s$ is defined as

$$\tilde{V}_{k,n}^s = \max_{k \in \mathcal{K}_n} \tilde{V}_k \quad (34)$$

The acquisition correlation values $\tilde{V}_{k,n}$ follow chi-square distribution of χ_{2N_c, λ_n}^2 , and the non-central parameters can be represented as

$$\lambda_n = N_c p_n^2 \frac{A^2}{\sigma^2} = 2p_n^2 N_c T_{\text{coh}} \cdot (C/N_0) \quad (35)$$

Taking $f(x; n, \lambda)$ to represent the PDF of $\chi_{n, \lambda}^2$, it can be known that (Shao 2003)

$$f(x; n, \lambda) = e^{-\frac{\lambda}{2}} \sum_{i=0}^{+\infty} \frac{1}{i!} \left(\frac{\lambda}{2} \right)^i \frac{x^{i+\frac{n}{2}-1}}{\Gamma\left(\frac{n}{2} + 1\right) 2^{i+\frac{n}{2}}} e^{-\frac{x}{2}} \quad (36)$$

where $\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt$ denotes the gamma function. However, the Cumulative Distribution Function (CDF) of $\chi^2_{n,\lambda}$, i.e. $F(x;n, \lambda) = \int_0^x f(v;n, \lambda) dv$, is difficult to express explicitly. Further, the PDF and CDF of $\tilde{V}_{k,n}^s$ can be obtained through the conclusions of order statistics (Shao 2003)

$$f_{\tilde{V}_{k,n}^s}(x) = N_n f(x; 2, \lambda_n) [F(x; 2, \lambda_n)]^{N_n-1} \quad F_{\tilde{V}_{k,n}^s}(x) = \int_0^x f_{\tilde{V}_{k,n}^s}(v) dv \quad (37)$$

Since $\tilde{V}_k^s$ represents the maximum of all acquisition value except the peak, it equals to $\max(\tilde{V}_k^s)$. Thus, the PDF of $\tilde{V}_k^s$ can be expressed as

$$f_{\tilde{V}_k^s}(x) = f_{\tilde{V}_{k,3}^s}(x) F_{\tilde{V}_{k,2}^s}(x) F_{\tilde{V}_{k,1}^s}(x) + F_{\tilde{V}_{k,3}^s}(x) (f_{\tilde{V}_{k,1}^s}(x) F_{\tilde{V}_{k,2}^s}(x) + F_{\tilde{V}_{k,1}^s}(x) f_{\tilde{V}_{k,2}^s}(x)) \quad (38)$$

By substituting (38) to (32), the relationship between r_A and C/N_0 can be obtained

$$r_A = \frac{2N_c + \lambda_0}{\int_0^{+\infty} x [f_{\tilde{V}_{k,3}^s}(x) F_{\tilde{V}_{k,2}^s}(x) F_{\tilde{V}_{k,1}^s}(x) + F_{\tilde{V}_{k,3}^s}(x) (f_{\tilde{V}_{k,1}^s}(x) F_{\tilde{V}_{k,2}^s}(x) + F_{\tilde{V}_{k,1}^s}(x) f_{\tilde{V}_{k,2}^s}(x))] dx} \quad (39)$$

The above equation is the third-order model for $\tilde{V}_k^s$, and it is a relatively accurate assumption. However, a fourth-order integral solution operation with $\Gamma(\cdot)$ is necessary, which requires a large amount of computation. Therefore, an appropriate model simplification is carried out.

Second-order model

Considering the difference between p_1 and p_2 is small, it is reasonable to assume that their effects are similar for $\tilde{V}_k^s$. Assuming $N_4 = N_1 + N_2$ represents the total number of non-ideal correlation peak p_1 and p_2 , $\tilde{V}_{k,4}^s$ represents the maximum values of N_4 independent identically distributed non-ideal correlation peaks. By substituting N_4 and $p_4 = p_1$ into (37), the PDF and CDF can be obtained. Furthermore, the PDF of the maximum value of $\tilde{V}_{k,3}^s$ and $\tilde{V}_{k,4}^s$ can be obtained as well

$$f_{\tilde{V}_{k,3}^s, \tilde{V}_{k,4}^s}(x) = f_{\tilde{V}_{k,3}^s}(x) F_{\tilde{V}_{k,4}^s}(x) + f_{\tilde{V}_{k,4}^s}(x) F_{\tilde{V}_{k,3}^s}(x) \quad (40)$$

Substituting (40) to (32), the relationship between r_A and C/N_0 under second-order hypothesis can be obtained

$$r_A = \frac{2N_c + \lambda_0}{\int_0^{+\infty} x (f_{\tilde{V}_{k,3}^s}(x) F_{\tilde{V}_{k,4}^s}(x) + f_{\tilde{V}_{k,4}^s}(x) F_{\tilde{V}_{k,3}^s}(x)) dx} \quad (41)$$

Equation (41) represents the second-order model which has lower complexity compared with (39).

First-order model

To decrease the computation further, the subsection proposes the first-order hypothesis model. The impact of p_3 is mapped to the adjustment of N_4 corresponding to p_4 . The acquisition correlation values corresponding to p_3 follow a central chi-square distribution $\chi^2_{2N_c, \lambda_3}$. The acquisition correlation values corresponding to p_4 follow a non-central chi-square distribution $\chi^2_{2N_c, \lambda_4}$, where λ_4 is given by (35). Further, the probability $P(\tilde{V}_{k,3}^s > \tilde{V}_{k,4}^s)$ under different C/N_0 conditions can be calculated by the following equation

$$P(\tilde{V}_{k,3}^s > \tilde{V}_{k,4}^s) = \int_0^{+\infty} (1 - F(x; 2, \lambda_3)) f(x; 2, \lambda_4) dx \quad (42)$$

The influence of $p_3 \approx 0$ relative to the influence of the non-ideal correlation peak p_4 increases as the probability in (42) increases. Then, $P(\tilde{V}_{k,3}^s > \tilde{V}_{k,4}^s)$ can be used to get $N_{4,a}$ which reflects the influence of p_3 as following (Ma et al. 2023)

$$N_{4,a} = 2P(\tilde{V}_{k,3}^s > \tilde{V}_{k,4}^s) N_3 + N_4 \quad (43)$$

Substituting the above equation into (37) and (32), it can be derived that

$$r_A = \frac{2N_c + \lambda_0}{\int_0^{+\infty} x [N_{4,a} f(x; 2, \lambda_4) (F(x; 2, \lambda_4))^{N_{4,a}-1}] dx} \quad (44)$$

Equation (44) represents the first-order model and it greatly clarifies the expression of r_A .

Summary

In summary, three hypothesis models for ACR method have been derived. These three models have decreasing computational complexity but inevitably result in a loss of estimation accuracy. Based on the properties of GPS L1 C/A code (Kaplan and Hegarty 2017), it is reasonable to set that $N_1 = 120$, $N_2 = 120$ and $N_3 = 782$. Monte Carlo simulations are conducted for these hypothesis models with $f_i = 4.13$ MHz and $f_s = 16.367$ MHz. Figure 3 shows the relationship between C/N_0 and r_A under different conditions

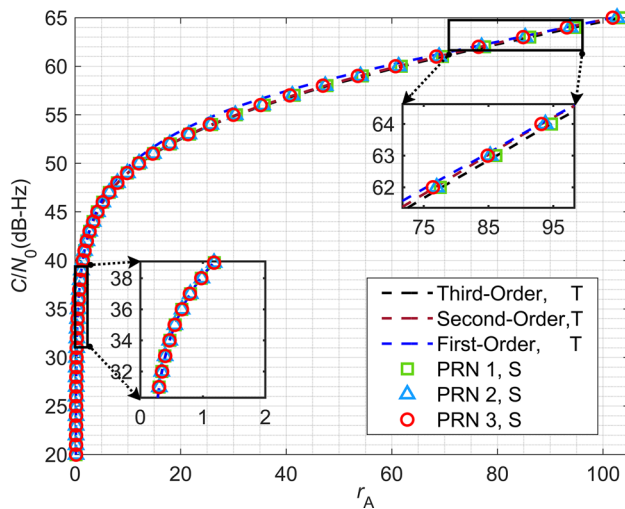


Fig. 3 Theoretical relationship between r_A and C/N_0 under different hypothesis models and simulation results for different PRNs

and C/N_0 can be estimated based on the corresponding r_A of signal.

It is shown that the third-order hypothesis model is the closest match to the simulation results. The estimation performance of the second-order model is slightly worse than the third-order model for high C/N_0 , but comparable for other C/N_0 values. As for the first-order model, it is clearly seen that there is a certain deviation between the simulation results and theoretical results. The reason is that the first-order model employs an approximate approach in the equivalent conversion to reduce the computational complexity. The maximum estimation error is approximately 0.6 dB-Hz, which is within an acceptable range for some practical applications.

In addition, the influence of different PRN sequences is small so it is reasonable to establish a unified model for all PRNs. Therefore, the choice of models depends on the desired computational complexity and estimation accuracy in practical applications. Further, the ACR method can apply to other GNSS constellations and the algorithm steps can be summarized as follows.

Step 1. Determine the autocorrelation function of satellites with estimated signals and analyze the properties to obtain p_n and N_n .

Step 2. Select the corresponding order model based on the expected estimation accuracy and computational complexity to obtain the relationship between r_A and C/N_0 .

Step 3. Calculate r_A with V_k^m and V_k^s of acquisition results in (31) and map it to get C/N_0 .

The cut-off C/N_0

The estimated Doppler frequency and code phase $(f_{d,k}, \tau_k)$ of the acquisition correlation peak $\tilde{V}_k^m$ possibly does not correspond to the true value (f_d, τ_a) of the signal shown in (24) owing to the influence of noise. And the probability becomes large with C/N_0 decreasing. So, the acquisition correlation ratio value corresponds to true (f_d, τ_a) is highly likely to be less than 1 but r_A defined by (31) is greater than 1 for low C/N_0 conditions. Once the non-ideal situation arises, the estimation result of ACR method will deviate and a large estimation error will occur.

Therefore, the cut-off carrier-to-noise ratio $(C/N_0)_T$ is defined as the lowest C/N_0 that makes the theoretical acquisition correlation ratio r_A greater than 1 in (32), and it is shown in the following equation (Ma et al. 2023)

$$(C/N_0)_T = \min\{C/N_0 | r_A = g(C/N_0) > 1\} \quad (45)$$

$(C/N_0)_T$ represents the range of C/N_0 which can be accurately estimated by ACR and it approximately equals to 38 dB-Hz when $T_{\text{coh}} = 1\text{ms}$ and $N_c = 1$. We can mainly increase T_{coh} to lower the $(C/N_0)_T$ if the acquisition of signal is not affected by the navigation bits such as CL code of GPS L2C which is not modulated by navigation data (Wu et al. 2023) and mainly increase N_c if signal is modulated by navigation data such as GPS L1 C/A code.

Performance evaluation

In this section, we derive theory performance of the proposed ACR method based on the third-order hypothesis model in detail and conduct Monte Carlo simulations. Owing to the complexity of the ACR expressions, we analyze performance through Padé approximation and cubic spline interpolation techniques. The simulation results show the impact of different factors like data length, coherent time and number of noncoherent integration on method performance.

Theory analysis

Assuming the numerator and denominator of (31) are denoted as X and Y

$$X = \frac{1}{N} \sum_{k=1}^N \tilde{V}_k^m \quad Y = \frac{1}{N} \sum_{k=1}^N \tilde{V}_k^s \quad (46)$$

The expectations of X and Y , i.e., μ_X and μ_Y , are represented as shown in (30). And their variances can be expressed as

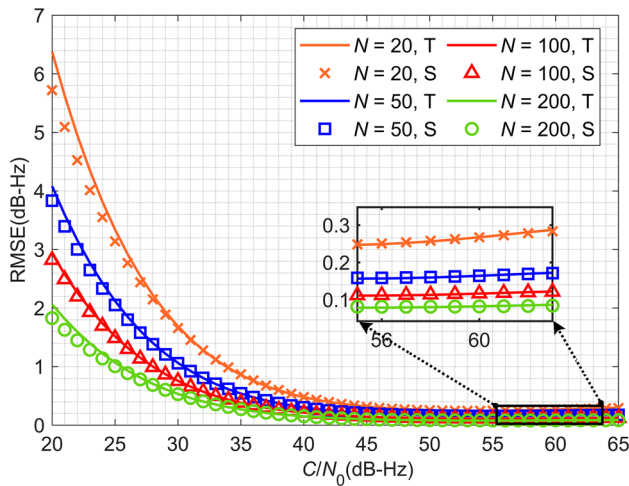


Fig. 4 C/N_0 estimation RMSE of the proposed ACR method under different data length N with coherent time $T_{\text{coh}} = 1$ ms and number of noncoherent integration $N_c = 1$

$$\sigma_X^2 = 2 \left(N_c + \frac{2A^2}{\sigma^2} \right) \quad \sigma_Y^2 = \int_0^{+\infty} (v - \mu_Y)^2 f_{\tilde{Y}^s}(v) dv \quad (47)$$

The implicit expressions for $E(r_A)$ and $\text{Var}(r_A)$ can be derived based on the rational fraction expectation and variance formulas (Kay 1993)

$$E\left(\frac{X}{Y}\right) = \frac{\mu_X}{\mu_Y} - \frac{\sigma_{XY}^2}{\mu_Y^2} + \frac{\mu_X}{\mu_Y^3} \sigma_Y^2 \quad (48)$$

$$\text{Var}\left(\frac{X}{Y}\right) = \left(\frac{\mu_X}{\mu_Y}\right)^2 \left(\frac{\sigma_X^2}{\mu_X^2} + \frac{\sigma_Y^2}{\mu_Y^2} - \frac{2\sigma_{XY}^2}{\mu_X \mu_Y} \right) \quad (49)$$

However, the lack of an explicit expression for $g(C/N_0)$ due to the complexity of the PDF of Y hinders the performance analysis of the ACR method. Direct computation of $E[(C/N_0)_A]$ and $\text{Var}[(C/N_0)_A]$ is extremely challenging because of the unknown functional form and the PDF of $g^{-1}(r_A)$.

To facilitate the computation, we approximate $g^{-1}(r_A)$ with another function. Considering that the high moments of the non-central chi-squared distribution do not have explicit expressions and the moments of high order rational fractions and power functions are also difficult to compute, we utilize a low-order rational fraction Padé approximation for $g^{-1}(r_A)$.

Padé approximation is a type of rational function approximation, which is more accurate than Taylor expansion for most functions (Baker and Gammel 1961). Assuming $f(r_A) = g^{-1}(r_A)$, when r_A is near $r_{A,0}$, it can be obtained that

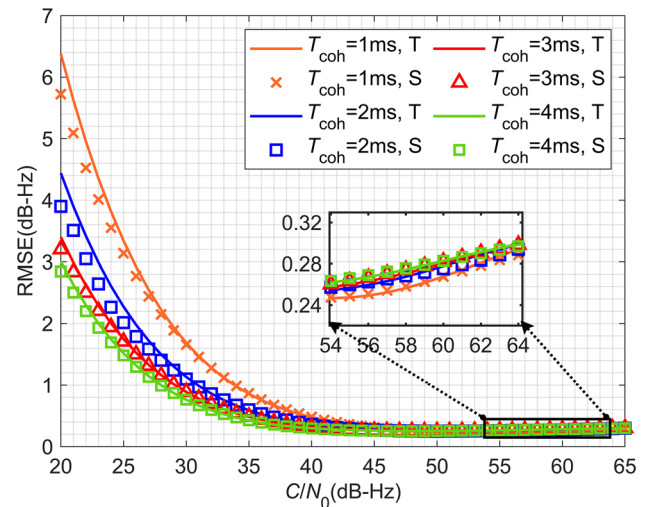


Fig. 5 C/N_0 estimation RMSE of the proposed ACR method under different coherent time T_{coh} with data length $N = 20$ and number of noncoherent integration $N_c = 1$

$$R_{1,1}(r_A) = \frac{a_0 + a_1 r_A}{1 + b_1 r_A} \approx f(r_{A,0})(r_A - r_{A,0}) + \frac{1}{2} f''(r_{A,0})(r_A - r_{A,0})^2 \quad (50)$$

The coefficient a_0 , a_1 and b_1 can be calculated from the above equation. Further, $E[(C/N_0)_A]$ and $\text{Var}[(C/N_0)_A]$ can be obtained by substituting X and Y in (48) and (49) with $X' = a_1 r_A + a_0$ and $Y' = b_1 r_A + 1$ (Basdevant 1972).

Further, we adopt the method of spline interpolation to solve the problem of calculating $f'(r_A)$ and $f''(r_A)$ when $f(r_A)$ is unknown. According to the properties of k th-degree spline interpolation function $S_k(x)$ (Shao 2003), a cubic spline interpolation satisfying the fixed, natural and periodic boundary conditions can be chosen to ensure the existence and continuity of the second derivative. The corresponding $f'(r_A)$ and $f''(r_A)$ at the point $r_A = r_{A,0}$ can be replaced by $S'_k(r_{A,0})$ and $S''_k(r_{A,0})$ (McKinley and Levine 1998).

In conclusion, by combining Padé approximation and cubic spline interpolation, the performance evaluation of the ACR method can be obtained by numerical calculation. The next subsection will provide numerical solutions and simulation results for the method performance.

Simulation analysis

Monte Carlo simulations are conducted with the immediate frequency $f_i = 4.13$ MHz and sampling rate $f_s = 16.367$ MHz. Figure 4 shows that as the data length increases, the RMSE of the estimation results gradually decreases. Figure 5 and Fig. 6 describe the theoretical and simulation results with

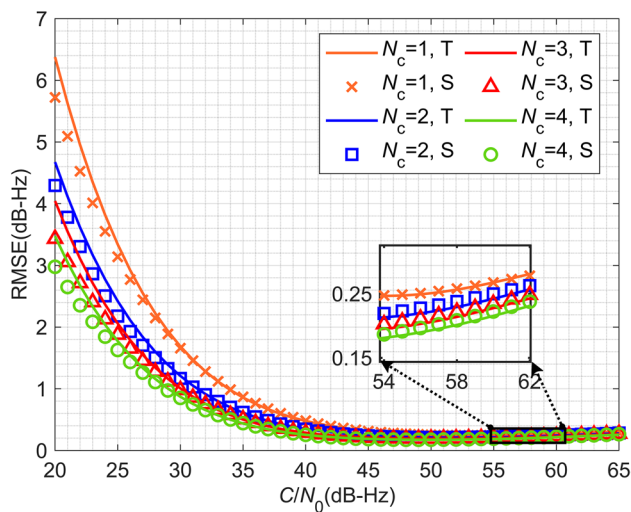


Fig. 6 C/N_0 estimation RMSE of the proposed ACR method under different number of noncoherent integration N_c with data length $N = 20$ and coherent time $T_{\text{coh}} = 1$ ms

different coherent time T_{coh} and number of noncoherent integration N_c , and the RMSE decreases as T_{coh} and N_c increase. The simulation results are basically consistent with the theoretical results, confirming the effectiveness of the theoretical analysis.

However, it should be noted that these conclusions do not hold true for the ACR method under high C/N_0 conditions and there are three unusual phenomena. Firstly, as shown in Fig. 4, the RMSE slightly increase for the same data length when C/N_0 is high. Secondly, the RMSE also exhibits a minor increase with an increase in T_{coh} and N_c when C/N_0 exceeds 53 dB-Hz in Fig. 5 and Fig. 6. Moreover, the RMSE of longer T_{coh} is higher while it is contrary for N_c under high C/N_0 conditions. These phenomena can be attributed to the following three straightforward explanations.

Firstly, the boundary effect matters. When C/N_0 is low, the theoretical value of r_A in (37) tends to zero. Meanwhile, the theoretical value of r_A will approach an upper limit, namely r_A^{up} , which is determined by the feature of auto-correlation function when C/N_0 is very high. Therefore, if the y-axis (C/N_0) range of Fig. 3 extends, the slope of the estimation curve will be very large as the x-axis (r_A) approaches r_A^{up} . Consequently, $(g^{-1}(r_A))$ will exhibit a decreasing trend followed by an increasing trend with r_A increasing from zero to r_A^{up} . Therefore, the absolute condition number which measures the ill-conditioning of solving function value problems (Shao 2003) for solving C/N_0 will decrease first and then increase as C/N_0 increases, which indirectly reflects the RMSE of estimation.

Secondly, the influence of hypothesis bias will be amplified as C/N_0 increases. The hypothesis assuming the same N_c for all PRNs does not perfectly matched with every satellite.

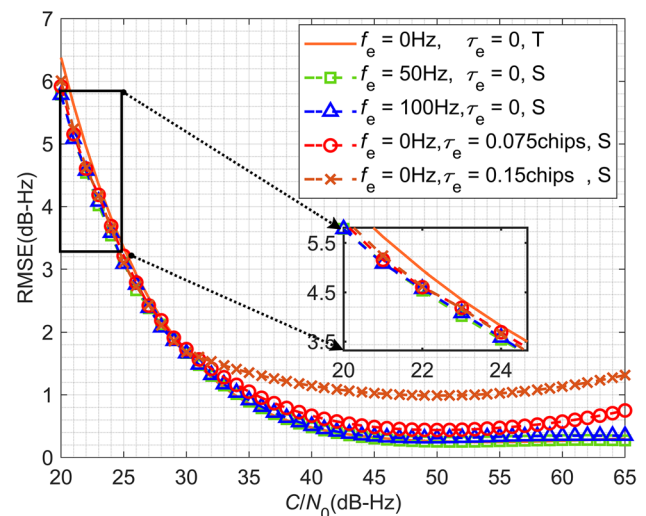


Fig. 7 C/N_0 estimation RMSE of the proposed ACR method under different acquisition frequency and phase errors with $N = 20$, $T_{\text{coh}} = 1$ ms and $N_c = 1$

It will become more biased since the impact of noise on the autocorrelation functions decreases under high C/N_0 conditions, exposing the deviation between the hypothesis and the actual autocorrelation function. Conversely, this error will be masked by the impact of noise when C/N_0 is relatively low. Increasing T_{coh} and N_c also reduces the influence of noise, thus amplifying this deviation under high C/N_0 conditions.

Thirdly, the impacts of coherent time T_{coh} and number of noncoherent integration N_c differ. T_{coh} only influences the noncentral parameters of the PDF of acquisition correlation values $V_k(f, \tau)$ while N_c is also related with the degrees of freedom except for noncentral parameters. For estimation under low C/N_0 conditions, the difficulty is to accurately estimate the signal power because the impact of noise for acquisition correlation values $V_k(f, \tau)$ is very significant, which is contrary to high C/N_0 conditions. Compared with noncoherent integration, coherent integration will lower the impact of noise to a greater extent, which is beneficial for low C/N_0 conditions while adverse for high C/N_0 conditions.

Figure 7 shows the impact of acquisition result errors on the ACR method where f_e , ϕ_e and τ_e denote the carrier frequency error, carrier phase error and code phase error respectively. If there is a certain error in the acquisition frequency, the RMSE of the ACR method will increase. When f_e reaches 100 Hz, the estimation results in a high C/N_0 environment will be approximately 0.1 dB-Hz increment which is within an acceptable error range. Based on (29), it can be known that the carrier phase does not affect the estimation. Code phase errors matter because a code phase error of 0.075 chips will worsen the result by 0.3 dB-Hz

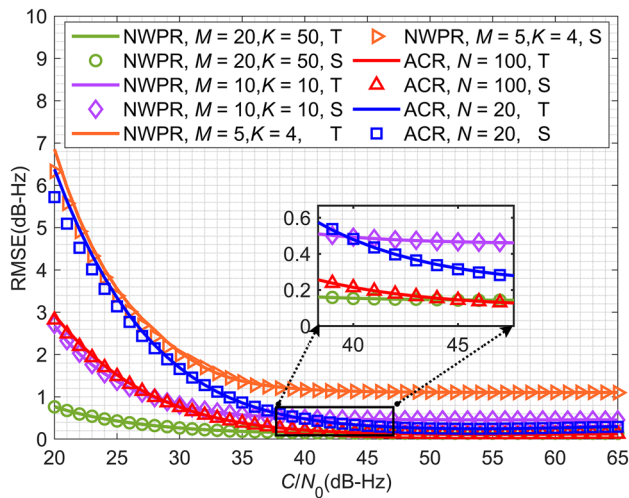


Fig. 8 Comparison of C/N_0 estimation RMSE for ACR and NWPR methods with $T_{\text{coh}} = 1$ ms and $N_c = 1$

Fig. 9 Experiment environment of signal simulator and real-world tests

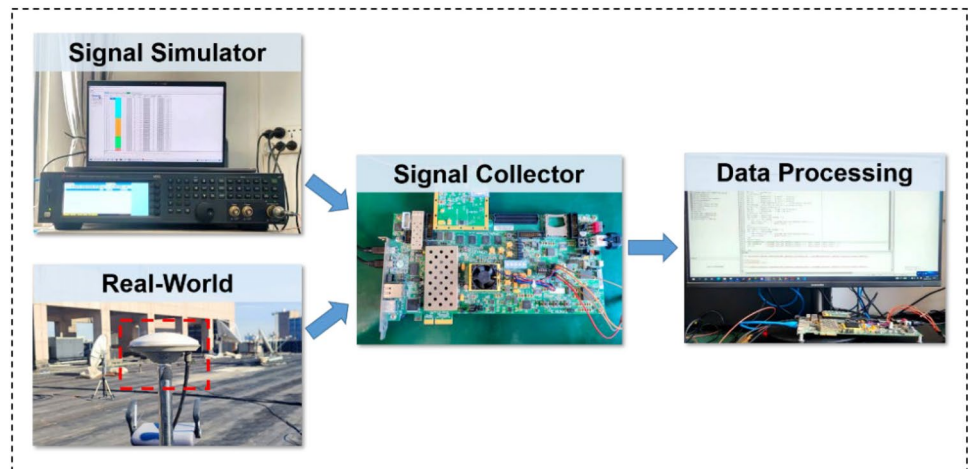


Table 1 C/N_0 estimation results of simulated signals with different power. The parameter of ACR is set to $N = 20$, $T_{\text{coh}} = 1$ ms, $N_c = 1$ and the parameters of NWPR-s and NWPR-i are set to $M = 5$, $K = 4$, $T_{\text{coh}} = 1$ ms

Power/dBm		− 106	− 111	− 116	− 121	− 126	− 131	− 136
Standard/dB-Hz		65.59	60.69	55.75	50.80	45.81	40.82	35.71
ACR /dB-Hz	Bias	− 0.15	0.26	0.31	0.04	0.18	0.44	0.52
	Std	0.23	0.21	0.35	0.26	0.35	0.52	0.73
	RMSE	0.27	0.33	0.47	0.27	0.40	0.68	0.90
NWPR-s /dB-Hz	Bias	− 0.14	0.13	− 0.30	− 0.31	0.07	− 0.04	− 0.28
	Std	1.08	1.15	1.37	1.25	1.19	1.32	1.59
	RMSE	1.09	1.16	1.40	1.29	1.19	1.32	1.61
NWPR-i /dB-Hz	Bias	− 0.55	− 0.34	− 0.33	− 0.98	0.13	− 0.07	− 1.57
	Std	1.29	1.08	1.51	1.18	1.78	1.65	2.27
	RMSE	1.40	1.13	1.55	1.53	1.79	1.66	2.76

on average. The limitation of acquisition code phase error can be solved in two ways. The first is to increase the sampling rate, which is easy to achieve for current commercial and specialized devices. The second way is interpolation between samples around the peak which is verified to have high accuracy (López-Risueno et al. 2005). It is worthwhile to trade a slight increase in computational complexity caused by high sampling rate or interpolation for an improvement in accuracy.

Finally, the estimation performance of the classical C/N_0 estimation methods are compared with ACR algorithm in Fig. 8. The results show the outperformance of ACR for normal and high C/N_0 conditions with the same data length as NWPR. However, due to the accuracy of acquisition correlation values and the impact of noise, the accuracy advantage of ACR under low C/N_0 is not significant. Besides, comparing ACR using 100 ms data with NWPR using 1,000 ms data, of which the parameters are $M = 20$, $K = 50$, it can be observed that the estimation performance of ACR with

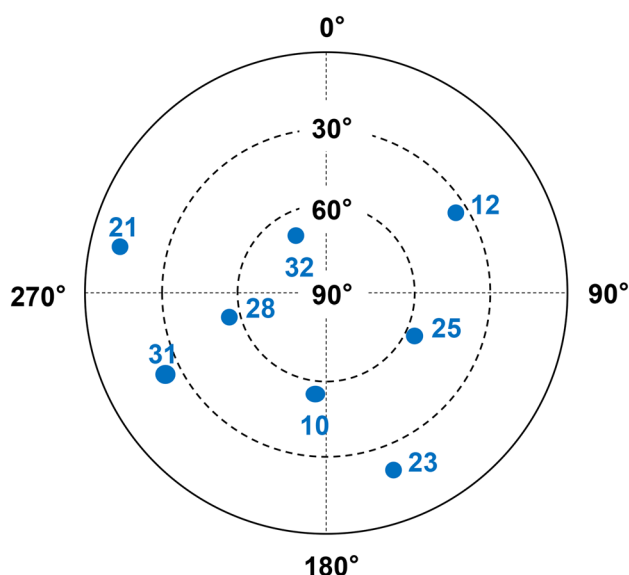


Fig. 10 Skyplot

Table 2 Parameters of the real-world experiment

Parameter	Settings
Time	2023–8–18 10:50 UTC
Position	N 40.0016° E 116.3301°
Satellites	10 12 21 23 25 28 31 32

nine tenth shorter data is even slightly better for C/N_0 higher than 45 dB-Hz. Therefore, the ACR method performs better than NWPR method for short data which we focus on in the experiment part to verify and it can achieve the approximate accuracy with shorter data when C/N_0 is high.

Experiment

To evaluate the feasibility of ACR method, two sets of experiments based on GPS L1 C/A code were conducted in two test cases where signals were generated from a GPS signal simulator and collected from the field. The signal collector was implemented using a self-developed receiver with the intermediate frequency $f_i = 46.42$ MHz and the sampling rate $f_s = 20$ MHz. Figure 9 shows the experiment environment of signal simulator and real-world tests. Considering that the NWPR method has the best performance among three classical methods, we collected 20 s signal to analyze the performance of ACR and NWPR methods for each test case.

Result of signal simulator test

GPS signals are generated at power levels ranging from -106dBm to -136dBm with a 5dBm interval using signal simulator. Considering the high accuracy of NWPR method with long data, the reference values, "Standard" in Table 1, are calculated by the NWPR method using 20 s data with the parameter settings $M = 20$, $K = 50$. "ACR" denotes the ACR estimation results. "NWPR-s" denotes the results obtained by using 20 ms I and Q integration values when tracking loop converges. "NWPR-i" denotes the C/N_0 calculated using the initial 20 ms I and Q integration values of the tracking loop, representing the estimation results of NWPR when there is only 20 ms data which is insufficient for the tracking loop to converge stably. "Bias" denotes the result difference and "Std" denotes the standard deviation.

Table 1 shows that the average estimation RMSE of ACR is lower than those of NWPR-s and NWPR-i and it achieves approximately 0.4 dB-Hz on average when C/N_0 exceeds 40 dB-Hz. The ACR results are slightly larger than its theoretical results, mainly due to the acquisition errors of carrier frequency and code phase. The standard deviation dominates the RMSE of NWPR-s because of its short data used for calculation. As for NWPR-i, almost all of the estimation biases are smaller than zero because the tracking loop is not stable in the early stage and the I and Q integration values fluctuate greatly, resulting in lower estimation results.

Result of real-world test

The real-world experiment is conducted and the estimation results of the ACR method are compared with the NWPR method. The experiment parameter settings are the same as the signal simulator experiment and the satellite information is depicted in Fig. 10 and Table 2.

The experiment results in Table 3 show that the average RMSE of ACR method is approximately 0.40 dB-Hz. The ACR method outperforms NWPR-i and NWPR-s, of which the RMSE exceed 1.3 dB-Hz and 2 dB-Hz respectively due to the instability of tracking loop initialization and short data. Especially, the RMSE of NWPR-i exceeds 7 dB-Hz for PRN 31. It is caused by the large acquisition error, which makes the tracking loop converge slowly and further leads to I and Q integration values with large errors in the initial stage. However, the ACR method estimation RMSE of PRN 31 is only 0.48 dB-Hz with the same acquisition results, which is consistent with the simulation results and analysis of acquisition errors in Fig. 7.

The experiment results are consistent with the theoretical and simulation results. Therefore, the ACR method can

Table 3 C/N_0 estimation results of real-world signals where the parameter settings are as same as Table 1

PRN		21	23	12	31	28	25	10	32
Standard/dB-Hz		35.97	38.13	42.80	44.64	46.55	46.83	47.16	51.70
ACR/dB-Hz	Bias	0.78	0.65	− 0.11	0.30	0.43	− 0.06	0.04	0.03
	Std	0.20	0.33	0.45	0.37	0.32	0.32	0.38	0.31
	RMSE	0.81	0.73	0.46	0.48	0.54	0.33	0.38	0.31
NWPR-s/dB-Hz	Bias	− 0.29	0.27	− 0.58	0.15	− 0.08	− 0.29	− 0.30	− 0.81
	Std	1.48	1.42	1.32	1.25	1.18	1.19	1.28	1.35
	RMSE	1.51	1.45	1.45	1.26	1.18	1.23	1.32	1.58
NWPR-i/dB-Hz	Bias	0.06	0.87	− 0.82	− 7.19	0.63	− 1.32	− 0.74	0.23
	Std	1.51	1.53	1.38	2.63	0.91	0.60	2.24	1.59
	RMSE	1.51	1.76	1.60	7.66	1.11	1.45	2.36	1.60

accurately estimate C/N_0 using short data in comparison with classical methods.

Conclusion

We proposed a C/N_0 estimation method based on the acquisition correlation ratio, i.e. the ACR method for short GNSS Data. And we elaborated three C/N_0 estimation methods and derived their theoretical performance which directly shows the unsatisfactory performance of classical methods under short data conditions. In addition, we deducted the theoretical performance of ACR method and compared it with classical methods, which gave a theoretical bound of ACR estimation results. Simulation and experiment were also conducted to verify the efficiency of ACR method.

The ACR method can provide a high-precision estimation of C/N_0 using a few milliseconds data in the acquisition stage. By constructing the relationship between the acquisition correlation ratio and C/N_0 , the ACR method addresses the limitation of classical methods under short data conditions and greatly improves the estimation precision. The theoretical RMSE of ACR method with 20 ms data is approximately 0.4 dB-Hz with C/N_0 larger than 40 dB-Hz, while the RMSE of NWPR method is more than 1 dB-Hz. Therefore, the ACR method can be applied to situations with sporadic, discontinuous or short data to provide an accurate estimation result of C/N_0 .

Appendix

The performance derivation of MM

Perform theoretical analysis on MM based on the Taylor expansion method (Kay 1993), and define that

$$\theta = \frac{\sqrt{2M_2^2 - M_4}}{M_2 - \sqrt{2M_2^2 - M_4}} \quad T = [M_2, M_4]^T \quad (51)$$

Let $Z_i = I_i^2 + Q_i^2$, then the expectation of M_2 and M_4 can be derived as follows

$$E(M_2) = E(Z_i) = A^2 + 2\sigma^2 \quad (52)$$

$$E(M_4) = E(Z_i^2) = A^4 + 8A^2\sigma^2 + 8\sigma^4 \quad (53)$$

Further, it can be obtained that

$$\left. \frac{\partial g}{\partial M_2} \right|_{T=\mu} = \frac{M_4}{\sqrt{2M_2^2 - M_4} (M_2 - \sqrt{2M_2^2 - M_4})^2} \bigg|_{T=\mu} = \frac{A^4 + 8A^2\sigma^2 + 8\sigma^4}{4A^2\sigma^4} \quad (54)$$

$$\left. \frac{\partial g}{\partial M_4} \right|_{T=\mu} = \frac{-M_2/2}{\sqrt{2M_2^2 - M_4} (M_2 - \sqrt{2M_2^2 - M_4})^2} \bigg|_{T=\mu} = \frac{A^2 + 2\sigma^2}{8A^2\sigma^4} \quad (55)$$

where the expectation $\mu = [E(M_2), E(M_4)]^T$. The elements of covariance matrix $C_T = [\text{Var}(M_2), \text{Cov}(M_2, M_4), \text{Cov}(M_2, M_4), \text{Var}(M_4)]$ can be expressed by the combination of $E(Z_i^n)$, $n = 1, 2, 3, 4$, and $E(Z_i^3)$ and $E(Z_i^4)$ can be derived by the properties of chi-square distribution

$$E(Z_i^3) = A^6 + 18A^4\sigma^2 + 72A^2\sigma^4 + 48\sigma^6 \quad (56)$$

$$E(Z_i^4) = A^8 + 32A^6\sigma^2 + 288A^4\sigma^4 + 768A^2\sigma^6 + 384\sigma^8 \quad (57)$$

Then, we can express C_T explicitly

$$C_T = \frac{1}{M} \begin{bmatrix} 4A^2\sigma^2 + 4\sigma^4 & 8A^4\sigma^2 + 48A^2\sigma^4 + 32\sigma^6 \\ 8A^4\sigma^2 + 48A^2\sigma^4 + 32\sigma^6 & 16A^6\sigma^2 + 208A^4\sigma^4 + 640A^2\sigma^6 + 320\sigma^8 \end{bmatrix} \quad (58)$$

By substituting (52) to (58) into (19), it can be obtained that

$$E[(C/N_0)_M] \approx C/N_0 \quad (59)$$

$$\begin{aligned} \text{Var}[(C/N_0)_M] \\ \approx \frac{1}{MT_{\text{coh}}^2} \left[\left(\frac{A^2}{\sigma^2} \right)^3 + \frac{33}{2} \left(\frac{A^2}{\sigma^2} \right)^2 + 92 \frac{A^2}{\sigma^2} \right. \\ \left. + 210 + 204 \left(\frac{A^2}{\sigma^2} \right)^{-1} + 68 \left(\frac{A^2}{\sigma^2} \right)^{-2} \right] \end{aligned} \quad (60)$$

The derivation of the PDF of $f_{\tilde{V}_k^s}(\mathbf{x})$

The CDF of $\tilde{V}_k^s$ in (38) of the third-order model can be derived as follow

$$F_{\tilde{V}_k^s}(x) = P(\tilde{V}_k^s < x) = P\left(\bigcap_{n=1}^3 (\tilde{V}_{k,n}^s < x)\right) = \prod_{n=1}^3 P(\tilde{V}_{k,n}^s < x) = \prod_{n=1}^3 F_{\tilde{V}_{k,n}^s}(x) \quad (61)$$

where the second equals sign is based on the definition of $\tilde{V}_{k,n}^s$, that is $\tilde{V}_{k,n}^s = \max_{k \in \mathcal{K}_n} \tilde{V}_k^s$. And the third equals sign is based on the independence of different correlation values $\tilde{V}_{k,n}^s$. Therefore, the PDF of $\tilde{V}_k^s$ can be obtained by taking the derivative of CDF

$$f_{\tilde{V}_k^s}(x) = \sum_{n=1}^3 \left(f_{\tilde{V}_{k,n}^s}(x) \prod_{q=1, q \neq n}^3 F_{\tilde{V}_{k,q}^s}(x) \right) \quad (62)$$

The $f_{\tilde{V}_k^s}(x)$ of the second-order model can also be derived similarly.

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Declarations

Competing interests The authors declare no competing interests.

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